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Fig 5.1 Our understanding of hormones, while still evolving, originated from a mix of random experiments, coincidental findings, and extraordinary sounding characters! One of these was the 'castrati' that featured in opera throughout the 16th, 17th, and 18th centuries. These were boys who were castrated before puberty. The voice of a castrato was pure and forceful, due to their enormous lung capacity and resulting breath control. They also experienced no temporal recession, and their arms and legs were long. One of the most well renowned was Farinelli, the stage name of Carlo Maria Michelangelo Nicola Broschi. It was said he had a well-modulated soprano voice with extraordinary breath control. His picture hangs in Handel's house in London. The practice was stopped in the early 20th century when it was acknowledged how inhumane the operation was.



The essence of endocrinology

For scientists

- Define a syndrome, and match it to a gland malfunction.
- Measure the gland's output in the peripheral blood. Define clinical syndromes associated with too much or too little secretion (*hyper-* and *hypo-*syndromes, respectively; *eu-* means normal, neither ↑ nor ↓, as in *euthyroid*). Note factors that may make measurement variable, eg diurnal release of cortisol.
- If suspecting hormone deficiency, test by stimulating the gland that produces it (eg short ACTH stimulation test or *Synacthen*® test in Addison's). If the gland is not functioning normally, there will be a blunted response to stimulation.
- If suspecting hormone excess, test by inhibiting the gland that produces it (eg dexamethasone suppression test in Cushing's). If there is a hormone-secreting tumour then this will fail to suppress via normal feedback mechanisms.
- Find a way to image the gland. NB: non-functioning tumours or 'incidentalomas' may be found in health, see p224. Imaging alone does not make the diagnosis.
- Aim to halt disease progression; diet and exercise can stop progression of impaired fasting glucose to frank diabetes.^{1,2} For other glands, halting progression will depend on understanding autoimmunity, and the interaction of genes and environment. In thyroid autoimmunity (an archetypal autoimmune disease), it is possible to track interactions between genes and environment (eg smoking and stress) via expression of immunologically active molecules (HLA class I and II, adhesion molecules, cytokines, CD40, and complement regulatory proteins).³

Endocrinologists love this reductionist approach, but have been less successful at understanding *emergent phenomena*—those properties and performances of ours that cannot be predicted from full knowledge of our perturbed parts. We understand the diurnal nature of cortisol secretion, for example, but the science of relating this to dreams, the consolidation of memory, and the psychopathology of families and other groups (such as the endocrinology ward round you may be about to join) is in its infancy. But as doctors we are steeped in the hormonal lives of patients (as they are in ours)—and we may as well start by recognizing this now.

For those doing exams

'What's wrong with *him*?' your examiner asks, boldly. While you apologize to the patient for this rudeness by asking, 'Is it alright if we speak about you as if you weren't here?', think to yourself that if you were a betting man or woman you would wager that the diagnosis will be endocrinological. In no other discipline are gestalt impressions so characteristic. To get good at recognizing these conditions, spend time in endocrinology outpatients and looking at collections of clinical photographs. Also, specific cutaneous signs are important, as follows.

Thyrotoxicosis: Hair loss; pretibial myxoedema (confusing term, p218); onycholysis (nail separation from the nailbed); bulging eyes (exophthalmos/proptosis).

Hypothyroidism: Hair loss; eyebrow loss; cold, pale skin; characteristic face. You might, perhaps should, fail your exam if you blurt out 'Toad-like face'.

Cushing's syndrome: Central obesity and wasted limbs (= 'lemon on sticks' see fig 5.2); moon face; buffalo hump; supraclavicular fat pads; striae.

Addison's disease: Hyperpigmentation (face, neck, palmar creases).

Acromegaly: Acral (distal) + soft tissue overgrowth; big jaws (macrognathia), hands and feet; the skin is thick; facial features are coarse.

Hyperandrogenism (♀): Hirsutism; temporal balding; acne.

Hypopituitarism: Pale or yellow tinged thinned skin, resulting in fine wrinkling around the eyes and mouth, making the patient look older.

Hypoparathyroidism: Dry, scaly, puffy skin; brittle nails; coarse hair.

Pseudohypoparathyroidism: Short stature, short neck, and short 4th and 5th metacarpals.



Fig 5.2 'Lemon on sticks.'

Hormones are chemical messengers which act directly on nearby cells (paracrine effect), on the cell of origin (autocrine effect), at a distant site (endocrine effect), or as neurotransmitters (brain and gastrointestinal tract). Thirst, thermal regulation, appetite, sleep cycles, menstrual cycle, and stress/mood are all controlled by the hypothalamus. Releasing factors produced by the hypothalamus reach the pituitary via the portal system (pituitary stalk), see **fig 5.3**. The releasing factors stimulate or inhibit the production of hormones from the anterior pituitary, **fig 5.4**. Vasopressin and oxytocin are produced in the hypothalamus and stored and released from the posterior pituitary.

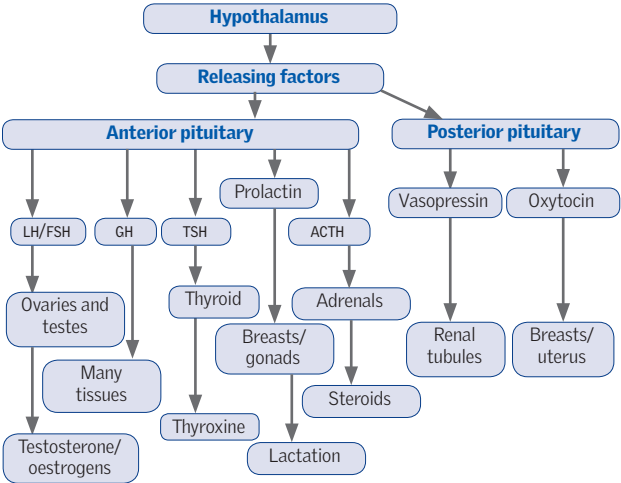


Fig 5.3 Hypothalamic-pituitary axis.

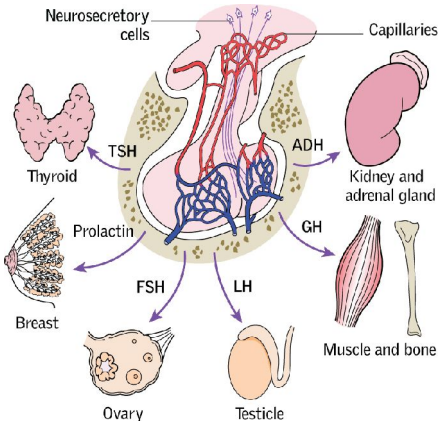


Fig 5.4 Neuroregulation and integration of endocrine axes makes us who we are—and who we are and what we do feeds back into our hormonal milieu. Multifactorial disruptions within the growth hormone (GH), luteinizing hormone (LH)–testosterone, adrenocorticotropin (ACTH)–cortisol and insulin axes play a major role in healthy maturation and ageing.



DM results from lack, or reduced effectiveness, of endogenous insulin. Hyperglycaemia is one aspect of a far-reaching metabolic derangement, which causes serious microvascular (retinopathy, nephropathy, neuropathy) or macrovascular problems: stroke, MI, renovascular disease, limb ischaemia. *So think of DM as a vascular disease:*¹ adopt a holistic approach and consider other cardiovascular risk factors too.

Categories of diabetes and dyslipidaemia

Type 1 DM: Usually adolescent onset but may occur at *any* age. *Cause:* insulin deficiency from autoimmune destruction of insulin-secreting pancreatic β cells. Patients must have insulin, and are prone to ketoacidosis and weight loss. Associated with other autoimmune diseases ($>90\%$ HLA DR3 $\pm$ DR4). Concordance is only $\sim 30\%$ in identical twins, indicating environmental influence. Four genes are important: one (6q) determines islet sensitivity to damage (eg from viruses or cross-reactivity from cows' milk-induced antibodies). Latent autoimmune diabetes of adults (LADA) is a form of type 1 DM, with slower progression to insulin dependence in later life.

Type 2 DM: (Formerly non-insulin-dependent DM, NIDDM) is at 'epidemic' levels in many places, mainly due to changes in lifestyle, but also because of better diagnosis and improved longevity.⁴ Higher prevalence in Asians, men, and the elderly (up to 18%). Most are over 40yrs, but teenagers are now getting type 2 DM (OHCS p156). *Cause:* $\downarrow$ insulin secretion $\pm$ $\uparrow$ insulin resistance. It is associated with obesity, lack of exercise, calorie and alcohol excess. $\geq 80\%$ concordance in identical twins, indicating stronger genetic influence than in type 1 DM. Typically progresses from a preliminary phase of impaired glucose tolerance (IGT) or impaired fasting glucose (IFG). (► This is a unique window for lifestyle intervention.) Maturity onset diabetes of the young (MODY) is a rare autosomal dominant form of type 2 DM affecting young people.

Impaired glucose tolerance (IGT): Fasting plasma glucose $<7\text{mmol/L}$ and OGTT (oral glucose tolerance) 2h glucose $\geq 7.8\text{mmol/L}$ but $<11.1\text{mmol/L}$.

Impaired fasting glucose (IFG): Fasting plasma glucose $\geq 6.1\text{mmol/L}$ but $<7\text{mmol/L}$ (WHO criteria). Do an OGTT to exclude DM. The cut-off point is somewhat arbitrary. IGT and IFG denote different abnormalities of glucose regulation (post-prandial and fasting). There may be lower risk of progression to DM in IFG than IGT. Manage both with lifestyle advice (p93) + annual review. Incidence of DM if IFG and HbA1c at high end of normal (37–46mmol/mol) is $\sim 25\%$.⁵

Other causes of DM ► Steroids; anti-HIV drugs; newer antipsychotics.

- Pancreatic: pancreatitis; surgery (where $>90\%$ pancreas is removed); trauma; pancreatic destruction (haemochromatosis, cystic fibrosis); pancreatic cancer.
- Cushing's disease; acromegaly; pheochromocytoma; hyperthyroidism; pregnancy.
- Others: congenital lipodystrophy; glycogen storage diseases.

Metabolic syndrome (syndrome X) Definition from International Diabetes Federation: central obesity (BMI >30 , or $\uparrow$ waist circ, ethnic-specific values), plus two of BP $\geq 130/85$, triglycerides $\geq 1.7\text{mmol/L}$, HDL $\leq 1.03\sigma/1.29\phi\text{mmol/L}$, fasting glucose $\geq 5.6\text{mmol/L}$ or type 2 DM. $\sim 20\%$ are affected; weight, genetics, and insulin resistance important in aetiology. Vascular events—but probably not beyond the combined effect of individual risk factors. R_x : Exercise; $\downarrow$ weight; treat individual components.

¹ Chicken or egg? Most type 2 diabetes-associated genes have a function in the vasculature, and stress in β -cells can result from vascular defects in the pancreas, so maybe vascular events trigger DM.⁴

Diagnosis of diabetes mellitus: WHO criteria (rather arbitrary!)

- Symptoms of hyperglycaemia (eg polyuria, polydipsia, unexplained weight loss, visual blurring, genital thrush, lethargy) AND raised venous glucose detected once—fasting ≥ 7 mmol/L or random ≥ 11.1 mmol/L OR
 - Raised venous glucose on two separate occasions—fasting ≥ 7 mmol/L, random ≥ 11.1 mmol/L or oral glucose tolerance test (OGTT)—2h value ≥ 11.1 mmol/L
 - HbA_{1c} ≥ 48 mmol/mol. Avoid in pregnancy, children, type 1 DM, and haemoglobinopathies.
- Whenever you have a needle in a vein, do a blood glucose (unless recently done); note if fasting or not. Non-systematic, but better than urine tests (false –ves).

Differentiating type 1 and 2 diabetes

Occasionally it may be difficult to differentiate whether a patient has type 1 or 2 DM, although they can present differently (see [table 5.1](#)). Features of type 1 include weight loss; persistent hyperglycaemia despite diet and medications; presence of autoantibodies: islet cell antibodies (ICA) and anti-glutamic acid decarboxylase (GAD) antibodies; ketonuria.

Table 5.1 Differences between type 1 and type 2 diabetes

	Type 1 DM	Type 2 DM
Epidemiology	Often starts before puberty	Older patients (usually)
Genetics	HLA D3 and D4 linked	No HLA association
Cause	Autoimmune β -cell destruction	Insulin resistance/ β -cell dysfunction
Presentation	Polydipsia, polyuria, $\downarrow$ weight, ketosis	Asymptomatic/complications, eg MI

► Not all new-onset DM in older people is type 2: if ketotic $\pm$ a poor response to oral hypoglycaemics (and patient is slim or has a family or personal history of autoimmunity), think of latent autoimmune diabetes in adults (LADA) and measure islet cell antibodies.

What is the best diet for obese patients with type 2 diabetes?

Dietary carbohydrate is a big determinant of postprandial glucose levels, and low-carbohydrate diets improve glycaemic control. How do low-carbohydrate, ketogenic diets (<20g of carbohydrate daily; LCKD) compare with low-glycaemic index, reduced-calorie diet (eg 500kcal/day deficit from weight maintenance diet)? In one randomized study over 24 weeks, LCKD had greater improvements in HbA_{1c} (–15 vs –5mmol/L), weight (–11kg vs –7kg), and HDL. Diabetes drugs were reduced or eliminated in 95% of LCKD vs 62% of LGID participants.⁶ NB: effects on renal function and mortality are unknown so these diets remain controversial.

Monitoring glucose control

- 1 Fingerprick glucose if on insulin (type 1/2). NB: *before* a meal informs about long-acting insulin doses; *after* meals inform about the dose of short-acting insulin.
- 2 Glycated haemoglobin (HbA_{1c}) relates to mean glucose level over previous 8wks (RBC t½). Targets are negotiable, eg 48–57mmol/mol (depends on patient's wish and arterial risk, eg past MI or stroke). If at risk from the effects of hypoglycaemia, eg elderly patients prone to falls, consider less tight control. Tight control may not alter all-cause mortality.⁷ Complications rise with rising HbA_{1c}, so any improvement helps.
- 3 Be sure to ask about hypoglycaemic attacks (and whether symptomatic). Hypoglycaemic awareness may diminish if control is too tight, or with time in type 1 DM, due to $\downarrow$ glucagon secretion. It may return if control is loosened.

General ▶ Focus on education and lifestyle advice (eg exercise to ↑ insulin sensitivity), healthy eating: ↓ saturated fats, ↓ sugar, ↑ starch-carbohydrate, moderate protein. Foods made just for diabetics are not needed. One could regard bariatric surgery as a cure for DM in selected patients. Be prepared to negotiate HbA_{1c} target and review every 3–6 months. Assess *global* vascular risk; start a high-intensity statin (p115), eg atorvastatin as tolerated, control BP (p211). Give foot-care (p212). (Pre-) pregnancy care should be in a multidisciplinary clinic (OHCS p23). Advise informing DVLA and not to drive if hypoglycaemic spells (p159; loss of hypoglycaemia awareness may lead to loss of licence; permanent if HGV).

Type 1 DM Insulin (see BOX 'Using insulin').

Type 2 DM See fig 5.5.

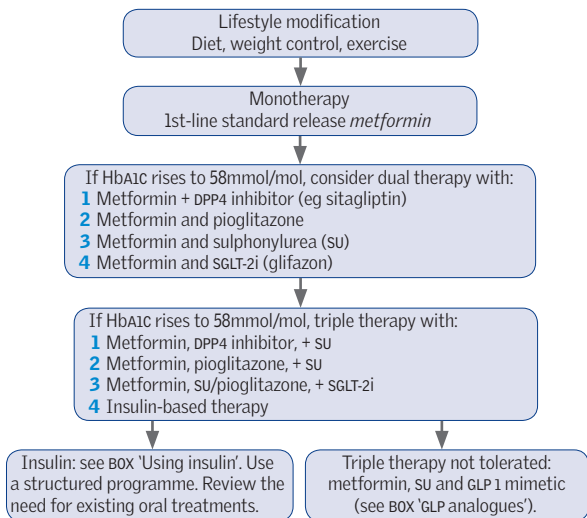


Fig 5.5 Management of type 2 diabetes. Aim for HbA_{1c} 48mmol/mol or 53 if two or more agents.

Data from Algorithm for blood glucose lowering therapy in adults with type 2 diabetes,

<http://www.nice.org.uk/guidance/ng28/chapter/1-Recommendations#drug-treatment-2>

Oral hypoglycaemic agents

Metformin: A biguanide. ↑ insulin sensitivity and helps weight. SE: nausea; diarrhoea (try modified-release version); abdominal pain; not hypoglycaemia. Avoid if eGFR ≤36mL/min (due to risk lactic acidosis).

DPP4 inhibitors/gliptins: (Eg sitagliptin.) Block the action of DPP-4, an enzyme which destroys the hormone incretin.

Glitazone: ↑ insulin sensitivity; SE: hypoglycaemia, fractures, fluid retention, ↑LFT (do LFT every 8wks for 1yr, stop if ALT up >3-fold). CI: past or present CCF; osteoporosis; monitor weight, and stop if ↑ or oedema.

Sulphonylurea: ↑ insulin secretion; eg gliclazide 40mg/d. SE: hypoglycaemia (monitor glucose); it ↑ weight.

SGLT: Selective sodium-glucose co-transporter-2 inhibitor. Blocks the reabsorption of glucose in the kidneys and promotes excretion of excess glucose in the urine (eg empagliflozin, shown to reduce mortality from cardiovascular disease in patients with type 2 DM, when compared to placebo).⁸

Using insulin

Vital to educate to self-adjust doses in the light of exercise, fingerprick glucose, calorie intake, and carbohydrate counting. •Phone support (trained nurse 7/24). •Can modify diet wisely and avoid binge drinking (danger of delayed hypoglycaemia). •Partner can abort hypoglycaemia: sugary drinks; GlucoGel® PO if coma (no risk of aspiration). •Dose titration to target—eg by 2–4 UNIT steps.

►► It is vital to write UNITS in full when prescribing insulin to avoid misinterpretation of U for zero!

Subcutaneous insulins Short-, medium-, or long-acting. Strength: 100u/mL.

- 1 Ultra-fast acting (Humalog®; Novorapid®); inject at start of meal, or just after (unless sugar-laden)—helps match what is actually eaten (vs what is planned).
- 2 Isophane insulin (variable peak at 4–12h): favoured by NICE (it's cheap!).
- 3 Pre-mixed insulins (eg NovoMix® 30 = 30% short-acting and 70% long-acting).
- 4 Long-acting recombinant human insulin analogues (*insulin glargine*) are used at bedtime in type 1 or 2 DM. There is no awkward peak, so good if nocturnal hypoglycaemia is an issue. Caution if considering pregnancy. *Insulin detemir* is similar and has a role in intensive insulin regimens for overweight type 2 DM.

Common insulin regimens ►Plan the regimen to suit the lifestyle, not vice versa. Disposable pens: dial dose; insert needle 90° to skin. Vary injection site (outer thigh/abdomen); change needle daily.

- 'BD biphasic regimen': twice daily premixed insulins by pen (eg NovoMix 30®)—useful in type 2 DM or type 1 with regular lifestyle.
- 'QDS regimen': before meals ultra-fast insulin + bedtime long-acting analogue: useful in type 1 DM for achieving a flexible lifestyle (eg for adjusting doses with size of meals, or exercise).
- Once-daily before-bed long-acting insulin: a good initial insulin regimen when switching from tablets in type 2 DM. Typical dose to work up to (slowly!): $\geq 1\text{u}/24\text{h}$ for every unit of BMI in adults. Consider retaining metformin ($\pm$ pioglitazone) if needed for tight control and patient is unable to use BD regimen.

Dose adjustment for normal eating (DAFNE): Multidisciplinary teams promoting autonomy can save lives. DAFNE found that training in flexible, intensive insulin dosing improved glycaemic control as well as well-being.⁹ It is resource intensive.

Subcutaneous insulin dosing during intercurrent illnesses (eg influenza)

- Advise patients to avoid stopping insulin during acute illness.
- Illness often increases insulin requirements despite reduced food intake.
 - Maintain calorie intake, eg using milk.
 - Check blood glucose ≥ 4 times a day and look for ketonuria. Increase insulin doses if glucose rising. Advise to get help from a specialist diabetes nurse or GP if concerned (esp. if glucose levels are rising or ketonuria). One option is 2-hourly ultra fast-acting insulin (eg 6–8u) preceded by a fingerprick glucose check.
 - Admit if vomiting, dehydrated, ketotic (►►p832), a child, or pregnant.

Insulin pumps (continuous subcutaneous insulin) Consider when attempts to reach HbA1c with multiple daily injections have resulted in disabling hypoglycaemia or person has been unable to achieve target HbA1c despite careful management.

Glucagon-like peptide (GLP) analogues (exenatide, liraglutide)

Work as incretin mimetics. Incretins are gut peptides that work by augmenting insulin release. Given by subcutaneous injection. Patients must have BMI >35 and specific psychological or other medical problems associated with obesity, or have a BMI lower than 35 kg/m^2 , and for whom insulin therapy would have significant occupational implications, or weight loss would benefit other significant obesity-related comorbidities. To continue a GLP 1 mimetic, a person should have a beneficial metabolic response (a reduction of HbA1c by at least 11mmol/mol) and a weight loss of at least 3% of initial body weight in 6 months.

Prospective studies show that good control of hyperglycaemia is key to preventing microvascular complications in type 1 and 2 DM.¹⁰ ► *Find out what problems are being experienced* (eg glycaemic control, morale, erectile dysfunction—p230).

Assess vascular risk BP control (see BOX 'Controlling blood pressure in diabetes') is crucial for preventing macrovascular disease and ↓mortality. Refer to *smoking* cessation services. Check *plasma lipids*.

Look for complications • Check injection sites for infection or lipohypertrophy (fatty change): advise on rotating sites of injection if present.

• **Vascular disease:** Chief cause of death. MI is 4-fold commoner in DM and is more likely to be 'silent'. Stroke is twice as common. Women are at high risk—DM removes the vascular advantage conferred by the female sex. Address other risk factors—diet, smoking, hypertension (p93). Suggest a statin (eg atorvastatin 20mg nocte) for all, even if no overt IHD, vascular disease, or microalbuminuria. Aspirin 75mg reduces vascular events (in context of secondary prevention). Safe to use in diabetic retinopathy.¹¹

• **Nephropathy:** (p314.) Microalbuminuria is when urine dipstick is -ve for protein but the urine albumin:creatinine ratio (UA:CR) is ≥3mg/mmol (units vary, check lab) reflecting early renal disease and ↑vascular risk. If UA:CR >3, inhibiting the renin-angiotensin system with an ACE-i or sartan, even if BP is normal, protects the kidneys. Spironolactone may also help.¹² Refer if UA:CR >7 ± GFR falling by >5mL/min/1.73m²/yr.¹³

• **Diabetic retinopathy:** Blindness is *preventable*. ► *Annual retinal screening mandatory for all patients*. Refer to an ophthalmologist if pre-proliferative changes or if any uncertainty at or near the macula (the only place capable of 6/6 vision). ► Pre-symptomatic screening enables laser photocoagulation to be used, aimed to stop production of angiogenic factors from the ischaemic retina. Indications: maculopathy or proliferative retinopathy. See **figs 5.6–5.9**.

• **Background retinopathy:** Microaneurysms (dots), haemorrhages (blots), and hard exudates (lipid deposits). Refer if near the macula, eg for intravitreal triamcinolone.

• **Pre-proliferative retinopathy:** Cotton-wool spots (eg infarcts), haemorrhages, venous beading. These are signs of retinal ischaemia. Refer to a specialist.

• **Proliferative retinopathy:** New vessels form. Needs urgent referral.

• **Maculopathy:** (Hard to see in early stages.) Suspect if ↓acuity. Prompt laser, intravitreal steroids, or anti-angiogenic agents may be needed in macular oedema. **Pathogenesis:** Capillary endothelial change → vascular leak → microaneurysms → capillary occlusion → local hypoxia + ischaemia → new vessel formation. High retinal blood flow caused by hyperglycaemia (and ↑BP and pregnancy) triggers this, causing capillary pericyte damage. Microvascular occlusion causes *cotton-wool spots* (± *blot haemorrhages* at interfaces with perfused retina). *New vessels* form on the disc or ischaemic areas, proliferate, bleed, fibrose, and can detach the retina. Aspirin² (2mg/kg/d) may be recommended by ophthalmologists; there is no evidence that it ↑bleeding.

• **Cataracts:** May be juvenile 'snowflake' form, or 'senile'—which occur earlier in diabetic subjects. Osmotic changes in the lens induced in acute hyperglycaemia reverse with normoglycaemia (so wait before buying glasses).

• **Rubeosis iridis:** New vessels on iris: occurs late and may lead to glaucoma.

• **Metabolic complications:** p832.

• **Diabetic feet:** p212.

• **Neuropathy:** p212.

² As DM has so many vascular events, particularly encourage statin use (p690), esp. if LDL >3mmol/L or systolic BP >140. Even consider a statin *whatever* the pre-treatment cholesterol; discuss with your patient.

Controlling blood pressure in diabetes

Type 1 DM: Treat BP if $>135/85\text{mmHg}$, unless albuminuria or two or more features of metabolic syndrome, in which case it should be $130/80\text{mmHg}$ (NICE 2015). Use an ACE-i 1st line or angiotension receptor antagonist if intolerant. If hypertensive and underlying renal involvement, see local guidance, p304.

Type 2 DM: Target BP $<140/80\text{mmHg}$ or $<130/80\text{mmHg}$ if kidney, eye, or cerebrovascular damage. 1st-line drug treatment should be an ACE-i, except in those of African or Caribbean origin, where ACE-i plus diuretic or a calcium-channel antagonist (CCA) should be started. For pregnant women offer CCA. Do not offer aspirin for the primary prevention of cardiovascular disease to adults with type 1 DM (NICE 2015). Don't combine an ACE-i with an angiotension receptor antagonist.



Fig 5.6 Background retinopathy, with microaneurysms and hard exudates.

Courtesy of Prof J Trobe.

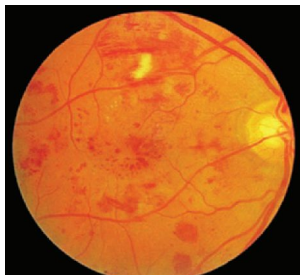


Fig 5.7 Pre-proliferative retinopathy, with haemorrhages and a cotton-wool spot.

Reproduced from Warrell *et al*, *Oxford Textbook of Medicine*, 2010, with permission from Oxford University Press.



Fig 5.8 Proliferative retinopathy, with new vessel formation and haemorrhages.

Courtesy of Prof J Trobe.



Fig 5.9 Scars from previous laser photocoagulation.

Courtesy of Prof J Trobe.

Improving quality of life: going beyond the pleasures of the flesh

'I cannot eat what I want because of your pitiful diet. Sex is out because diabetes has made me impotent. Smoking is banned, so what's left? I'd shoot myself if only I could see straight.' Start by acknowledging your patient's distress. Don't shrug it off—but don't take it at face value either. Life may be transformed by cataract surgery, sildenafil (unless contraindicated, p230), dietary negotiation, and sport (it needn't be shooting). Take steps to simplify care. Stop blood glucose self-monitoring if it's achieving nothing (known to ↓ quality of life). Even if all these interventions fail, you have one trump card up your sleeve: 'Let's both try to find one new thing of value before we next meet—and compare notes'. This opens the way to vicarious pleasure: a whole new world.

► Refer *early* to foot services (podiatry, imaging, vascular surgery). Amputations are common (135/week)—and preventable: *good care saves legs*. Examine feet regularly. Distinguish between ischaemia (critical toes ± absent foot pulses and worse outcome) and peripheral neuropathy (injury or infection over pressure points, eg the metatarsal heads). In practice, many have both.

Neuropathy ↓ Sensation in 'stocking' distribution: test sensation with a 10g monofilament fibre (sensory loss is patchy so examine all areas), absent ankle jerks, neuropathic deformity (Charcot joint, **fig 5.11**): pes cavus, claw toes, loss of transverse arch, rocker-bottom sole. Caused by loss of pain sensation, leading to mechanical stress and repeated joint injury. Swelling, instability, and deformity. ► Early recognition is vital (cellulitis or osteomyelitis are often misdiagnosed).

Ischaemia If the foot pulses cannot be felt, do Doppler pressure measurements. Any evidence of neuropathy or vascular disease raises risk of foot ulceration. *Educate* (daily foot inspection—eg with a mirror for the sole; comfortable shoes). *Regular chiropody* to remove callus, as haemorrhage and tissue necrosis may occur below, leading to ulceration. *Treat fungal infections* (p408). *Surgery* (including endovascular angioplasty balloons, stents, and subintimal recanalization) has a role.

Foot ulceration Typically painless, punched-out ulcer (**fig 5.10**) in an area of thick callus ± superadded infection. Causes cellulitis, abscess ± osteomyelitis.

Assess degree of:

- 1 Neuropathy (clinically).
- 2 Ischaemia (clinically + Doppler ± angiography).
- 3 Bony deformity, eg Charcot joint (clinically + x-ray). See **fig 5.11**.
- 4 Infection (swabs, blood culture, x-ray for osteomyelitis, probe ulcer to reveal depth).

Management: Regular chiropody. Bed rest ± therapeutic shoes. For Charcot joints: bed rest/crutches/total contact cast until oedema and local warmth reduce and bony repair is complete (≈8wks). Bisphosphonates may help. Charcot joints are also seen in tabes dorsalis, spina bifida, syringomyelia, and leprosy. Metatarsal head surgery may be needed. If there is cellulitis, admit for IV antibiotics. Common organisms: staphs, streps, anaerobes. Start empirically (as per your local guidance) with benzylpenicillin 1.2g/6h IV and flucloxacillin 1g/6h IV ± metronidazole 500mg/8h IV. IV insulin may improve healing. Get surgical help early. The degree of peripheral vascular disease, general health, and patient request will determine degree of vascular reconstruction/surgery.

► **Absolute indications for surgery:** Abscess or deep infection; spreading anaerobic infection; gangrene/rest pain; suppurative arthritis.

Diabetic neuropathies *Symmetric sensory polyneuropathy:* ('glove & stocking' numbness, tingling, and pain, eg worse at night). *R:* paracetamol → tricyclic (amitriptyline 10–25mg nocte; gradually ↑ to 150mg) → duloxetine, gabapentin, or pregabalin → opiates. Avoiding weight-bearing helps. *Mononeuritis multiplex:* (eg III & VI cranial nerves). Treatment: hard! If sudden or severe, immunosuppression may help (corticosteroids, IV immunoglobulin, ciclosporin). *Amyotrophy:* Painful wasting of quadriceps and other pelfemoral muscles. Use electrophysiology to show, eg lumbosacral radiculopathy, plexopathy, or proximal crural neuropathy. Natural course: variable with gradual but often incomplete improvement. IV immunoglobulins have been used. *Autonomic neuropathy:* (p505) Postural BP drop; ↓cerebrovascular autoregulation; loss of respiratory sinus arrhythmia (vagal neuropathy); gastroparesis; urine retention; erectile dysfunction; gustatory sweating; diarrhoea (may respond to codeine phosphate). Gastroparesis (early satiety, post-prandial bloating, nausea/vomiting) is diagnosed by gastric scintigraphy with a ⁹⁹technetium-labelled meal; anti-emetics, erythromycin, or gastric pacing. Postural hypotension may respond to fludrocortisone (SE: oedema, ↑BP)/midodrine (α-agonist; SE: ↑BP).



Preventing loss of limbs: primary or secondary prevention?

Traditionally prevention involves foot care advice in diabetic clinics (eg 'Don't go bare-foot'), promoting euglycaemia and normotension. But despite this, the sight of a diabetic patient minus one limb is not rare, and must prompt us to redouble our commitment to primary prevention, ie stopping those at risk from ever getting diabetes. The sequelae of diabetic neuropathy can lead to gangrene, amputation, and the impact on quality of life can be profound. As one patient post amputation said, 'I begin again to walk, on crutches. What nuisance, what fatigue, what sadness, when I think about all my ancient travels, and how active I was just 5 months ago! Where are the runnings across mountains, the walks, the deserts, the rivers, and the seas? And now, the life of a legless cripple. For I begin to understand that crutches, wooden and articulated legs, are a pack of jokes...Goodbye to family, goodbye to future! My life is gone, I'm no more than an immobile trunk' (Arthur Rimbaud. Letter to his sister Isabelle, 10 July 1891).



Fig 5.10 Gangrene (toes 2, 4, and 5).

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Fig 5.11 Charcot (neuropathic) joint.

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Special situations in diabetes

Pregnancy: (OHCS p23) 4% are complicated by DM: either pre-existing (<0.5%), or new-onset gestational diabetes (GDM) (>3.5%).

- All forms carry an increased risk to mother and fetus: miscarriage, pre-term labour, pre-eclampsia, congenital malformations, macrosomia, and a worsening of diabetic complications, eg retinopathy, nephropathy.
- Risk of GDM ↑ if: aged over 25; family history; +ve; tweight; non-Caucasian; HIV+ve; previous gestational DM.
- Pre-conception: offer general advice, and discuss risks. Control/reduce weight, aim for good glucose control, offer folic acid 5mg/d until 12 weeks.
- Screen for GDM with OGTT if risk factors at booking (16–18 weeks if previous GDM).
- Oral hypoglycaemics other than metformin should be discontinued. Metformin may be used as an adjunct or alternative to insulin in type 2 DM or GDM.

6wks postpartum, do a fasting glucose. Even if –ve, 50% go on to develop DM.

Surgery: Optimal blood sugar control pre, peri, and post operatively is important to minimize risk of infection and balance catabolic response to surgery. Type 1 diabetics should ideally be first on the list and BMs should have been stabilized 1–2 days pre major surgery. Consult local policy for how to manage insulin-treated/non-insulin-treated patients on morning of surgery (eg setting up glucose/insulin infusion).

Acute illness: Diabetics are prone to hyperglycaemia during periods of illness, in spite of reduced oral intake. Avoid stopping insulin in periods of acute illness.

►► **Hypoglycaemia** Commonest endocrine emergency—see p834. Prompt diagnosis and treatment essential—brain damage & death can occur if severe or prolonged.

Definition Plasma glucose $\leq 3\text{mmol/L}$. Threshold for symptoms varies. See BOX.

Symptoms *Autonomic*: Sweating, anxiety, hunger, tremor, palpitations, dizziness.

Neuroglycopenic: Confusion, drowsiness, visual trouble, seizures, coma. Rarely focal symptoms, eg transient hemiplegia. Mutism, personality change, restlessness, and incoherence may lead to misdiagnosis of alcohol intoxication or even psychosis.

Fasting hypoglycaemia *Causes*: The chief cause is insulin or sulfonylurea treatment in a diabetic, eg $\uparrow$ activity, missed meal, accidental or non-accidental overdose (check for circulating oral hypoglycaemics). In *non-diabetics* you must **EXPLAIN** mechanism: **E**xogenous drugs, eg insulin, oral hypoglycaemics (p208)? access through diabetic in the family? Body-builders may misuse insulin to help stamina. Also: alcohol, eg a binge with no food; aspirin poisoning; ACE-i; β -blockers; pentamidine; quinine sulfate; aminoglutethamide; insulin-like growth factor.¹⁴

Pituitary insufficiency.

Liver failure, plus some rare inherited enzyme defects.

Addison's disease.

Islet cell tumours (insulinoma) and immune hypoglycaemia (eg anti-insulin receptor antibodies in Hodgkin's disease).

Non-pancreatic neoplasms, eg fibrosarcomas and haemangiopericytomas.

When to investigate:

- Whipple answered this (*Whipple's triad*): symptoms or signs of hypoglycaemia + $\downarrow$ plasma glucose + resolution of symptoms or signs post glucose rise.
- Document BM during attack and lab glucose if in hospital (monitors often not reliable at low readings).
- Take a drug history and exclude liver failure.
- 72h fasting may be needed (monitor closely). Bloods: glucose, insulin, c-peptide, and plasma ketones if symptomatic. If endogenous hyperinsulinism suspected, do insulin, c-peptide, proinsulin, β -hydroxybutyrate.

Interpreting results:

- Hypoglycaemic hyperinsulinaemia (HH): *Causes*: insulinoma, sulfonylureas, insulin injection (no detectable c-peptide—only released with endogenous insulin); non-insulinoma pancreatogenous hypoglycaemia syndrome, mutation in the insulin-receptor gene. Congenital HH follows mutations in genes involved in insulin secretion (ABCC8, KCNJ11, GLUD1, CGK, HADH, SLC16A1, HNF4A, ABCC8, & KCNJ11).¹⁵
- Insulin low or undetectable, no excess ketones. *Causes*: non-pancreatic neoplasm; anti-insulin receptor antibodies.
- $\downarrow$ Insulin, $\uparrow$ ketones. *Causes*: alcohol, pituitary insufficiency, Addison's disease.

Post-prandial hypoglycaemia May occur after gastric/bariatric surgery ('dumping', p623), and in type 2 DM. *Investigation*: Prolonged OGTT (5h, p206).

Treatment ►► See p834. If episodes are often, advise many small high-starch meals. If post-prandial $\downarrow$ glucose, give slowly absorbed carbohydrate (high fibre). In diabetics, rationalize insulin therapy (p209).

The definition of hypoglycaemia is context-dependent

The brain stops working if plasma glucose levels get too low, so we are nervous of levels $\leq 3\text{mmol/L}$. But some are asymptomatic at this level. So what is definitely abnormal? The answer may be 4mmol/L , allowing for inaccuracies in fingerprick BMs (NB: whole blood glucose is 10–15% < plasma glucose.) Think: '*In this ill patient when can I be sure that a low glucose is not contributing to their illness?*' If $<4\text{mmol/L}$, you may be wise to treat (p834)—just in case. Consider, is the patient on hypoglycaemics, have they binged on alcohol 24hrs pre test? Skipped meals? Is there an underlying illness, eg insulinoma? Unlikely, but possible. Keep an open mind; let the GP know. Counsel patient and relative about warning signs of hypoglycaemia. Be more inclined to investigate if the effects of even mild hypoglycaemia might be disastrous (eg in pilots) or if there are unexplained symptoms.

Insulinoma

This often benign (90–95%) pancreatic islet cell tumour is sporadic or seen with MEN-1 (p223). It presents as fasting hypoglycaemia, with Whipple's triad:

- 1 Symptoms associated with fasting or exercise.
- 2 Recorded hypoglycaemia with symptoms.
- 3 Symptoms relieved with glucose.

Screening test Hypoglycaemia + ↑plasma insulin during a long fast.

Suppressive tests Give IV insulin and measure C-peptide. Normally exogenous insulin suppresses C-peptide production, but this does not occur in insulinoma.

Imaging CT/MRI ± endoscopic pancreatic US ± IACS (see Box; all fallible, so don't waste too much time before proceeding to intra-operative visualization ± intra-operative ultrasound). ¹⁸F-L-3,4-dihydroxyphenylalanine PET-CT can help guide laparoscopic surgery.

Treatment Excision.

Nesidioblastosis See Box. If this doesn't work, options are: diet, diazoxide, dextrose IV, enteral feeding, everolimus.

▶ **Diabetic ketoacidosis** Results from insulin deficiency (eg unknown diagnosis, intercurrent illness, interruption of insulin therapy). See p832.

Pursuing a voyage to the islets of Langerhans to the bitter end

A 50-year-old had episodic early-morning sweats and tremors and was found to have hyperinsulinaemic hypoglycaemia (= Nesidioblastosis). Selective intra-arterial calcium infusions (IACS) showed a 2-fold increase in insulin secretion after infusion of the splenic and superior mesenteric arteries, so setting the stage for 'hunt the insulinoma'.

But cross-sectional imaging and endoscopic ultrasound were normal. At laparotomy, no lesion was found despite mobilization of the pancreas, or during intra-operative ultrasound. 'Time to sew up and go home?' 'No!' said the surgeon, 'I'm going to do a distal pancreatectomy'. Histology showed no discrete insulinoma, but diffuse islet cell hyperplasia (*nesidioblastosis*). How much pancreas to resect? Too little and nothing is gained: too much spells pancreatic endocrine disaster. Luckily the surgeon guessed right, and the patient was cured by the procedure.¹⁶

